

Curated Research Articles

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- **Novel High-Temperature Chemistry in All-NASICON Solid State Batteries Explored by Operando Synchrotron XRD** — score: 1.000 The article investigates the high-temperature chemical processes occurring in all-NASICON solid-state batteries using operando synchrotron X-ray diffraction (XRD). The study aims to enhance the understanding of structural changes and material stability during battery operation, contributing to advancements in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Weakly solvated Li⁺ chemistry within MXene nanochannels enables wide-temperature 2-electron oxalate-routine Li-CO₂ battery** — score: 1.000 The article discusses a novel approach to enhance the performance of Li-CO₂ batteries by utilizing weakly solvated Li⁺ ions within MXene nanochannels. This method allows the battery to operate efficiently across a wide temperature range, potentially addressing the challenges of energy storage and conversion in lithium-based systems. Journal: *ScienceDirect Publication: Nano Energy*
- **Rapid impedance spectrum extraction from relaxation signals via a battery kinetics-informed neural network** — score: 0.800 The article presents a novel approach for quickly extracting impedance spectra from relaxation signals using a neural network informed by battery kinetics. This method aims to enhance the efficiency of analyzing battery performance, potentially leading to faster advancements in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Unveiling the asymmetry in sodium-ion charge-discharge behavior via engineered internal vacancy defects in closed pores** — score: 0.800 The article explores how engineered internal vacancy defects within closed pores can create asymmetry in the charge-discharge behavior of sodium-ion batteries, potentially enhancing their performance. This study aims to optimize sodium-ion technology by revealing the impact of structural modifications on battery efficiency. Journal: *ScienceDirect Publication: Nano Energy*
- **Scalable interface-interphase design of tin electrodes enables pouch-type sodium all-solid-state batteries** — score: 0.800 The article introduces an innovative interface-interphase engineering strategy for tin electrodes in sodium all-solid-state batteries (NaSSBs). This method, compatible with roll-to-roll manufacturing, incorporates a sodium closo-borate anolyte within a carbon nanotube scaffold, facilitating reversible alloying and low-pressure operations, ultimately leading to the successful demonstration of the first pouch-type NaSSB. Journal: *Joule*
- **Hard-Soft Acid-Base Theory-Guided Molecular Engineering for Synergistic Bulk-to-Interface Regulation in Ultra-Stable Aqueous Zinc-Ion Batteries** — score: 0.800 The article discusses the application of Hard-Soft Acid-Base (HSAB) theory to enhance the design of aqueous zinc-ion batteries, focusing on achieving synergistic regulation from bulk properties to interface interactions. This molecular engineering approach aims to improve the stability and performance of these batteries, which are increasingly important for energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **In situ Raman diagnostics of gas evolution during overcharge-induced failure in Li-ion batteries** — score: 0.700 The article discusses the use of in situ Raman spectroscopy to analyze gas emissions from Li-ion batteries during overcharging, which can lead to failure. This diagnostic method provides insights into the electrochemical processes and potential hazards associated with battery overcharge conditions, contributing to improved safety and performance in energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Processing-Induced Heterogeneity Governs Failure in Oxide Solid Electrolytes for Solid-State Batteries** — score: 0.700 The article discusses how heterogeneity induced by processing methods affects the performance and failure mechanisms of oxide solid electrolytes in solid-state batteries. It emphasizes the critical role of material processing in determining the reliability and efficiency of these

batteries, suggesting that a better understanding of this relationship is essential for future advancements in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Switching from insertion to conversion for multielectron aqueous vanadium batteries** — score: 0.600 The article discusses a significant advancement in aqueous vanadium-ion batteries by shifting from a one-electron insertion mechanism to a four-electron conversion reaction, which leads to higher charge storage capacities. This improvement is achieved through the application of a mesoporous carbon shell, marking a potential breakthrough in battery technology. Journal: *Nature Materials*
- **Widening electrochemical window of ether electrolyte via solvent-diluent intermolecular interactions for high-voltage lithium metal batteries** — score: 0.600 The article discusses a novel approach to enhance the electrochemical stability of ether electrolytes by manipulating solvent-diluent intermolecular interactions, which significantly widens the electrochemical window. This advancement aims to improve the performance of high-voltage lithium metal batteries, potentially contributing to more efficient energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Vertically aligned h-BN array enabling inorganic-rich SEI layer generation in polymer electrolyte towards ultralong lifespan solid-state lithium metal battery** — score: 0.600 The article presents a novel vertically aligned h-BN (hexagonal boron nitride) array designed to enhance the stability of solid-state lithium metal batteries by promoting the formation of an inorganic-rich solid electrolyte interphase (SEI) layer within a polymer electrolyte. This advancement aims to significantly extend the lifespan of these batteries, positioning them as a more reliable energy storage solution. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Regulating anion participation in phosphate electrolyte for nonflammable and low-temperature sodium-ion batteries** — score: 0.600 The article discusses the development of a phosphate electrolyte that regulates anion participation, aimed at enhancing the performance of nonflammable and low-temperature sodium-ion batteries. Key findings include improvements in battery safety and efficiency, which could significantly impact the future of energy storage technologies. Journal: *ScienceDirect Publication: Nano Energy*
- **Regenerating degraded polycrystalline NCM towards high-performance single-crystal NCM622 through tailoring crystal-phase reconstruction** — score: 0.600 The article discusses a method for improving the performance of degraded polycrystalline NCM (Nickel Cobalt Manganese) materials by converting them into high-performance single-crystal NCM622 through a process of crystal-phase reconstruction. This innovative approach aims to enhance the material's efficiency for energy applications, as detailed in findings presented in Nano Energy. Journal: *ScienceDirect Publication: Nano Energy*
- **Decoordination-enabled dynamic Lewis acid regulation in poly (ethylene oxide) solid polymer electrolytes for high-performance all-solid-state lithium metal batteries** — score: 0.500 The article discusses advancements in poly(ethylene oxide) solid polymer electrolytes, specifically focusing on how decoordination mechanisms can enhance the regulation of Lewis acids, leading to improved performance in all-solid-state lithium metal batteries. This innovation aims to address common challenges in energy storage materials, paving the way for more efficient battery technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Surface passivation and structural stabilization of Ni-rich and Co-free single-crystal cathodes to improve high-voltage performance in sulfide solid-state batteries** — score: 0.500 The article discusses advancements in enhancing the performance of Ni-rich and Co-free single-crystal cathodes for sulfide solid-state batteries by applying surface passivation and structural stabilization techniques. These methods aim to improve high-voltage capabilities, thus potentially increasing the efficiency and longevity of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Competitive adsorption-driven anion redistribution enables kinetics-stability decoupling in alkaline Zn-Fe flow batteries** — score: 0.500 The article discusses a novel approach in alkaline Zn-Fe flow batteries that utilizes competitive adsorption to facilitate the redistribution of anions,

thereby achieving a decoupling of reaction kinetics from stability. This advancement could lead to improved performance and longevity of battery systems, as outlined in the September 2026 publication in *Energy Storage Materials*. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Site-selective cation substitution enables ultra-fast charging and long stability in NASICON sodium-ion anodes** — score: 0.400 The article discusses a novel approach to enhancing sodium-ion battery anodes through site-selective cation substitution in NASICON materials, resulting in significantly improved charging speeds and extended stability. This advancement could play a crucial role in the development of more efficient and durable energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Porosity-gradient dry-processed graphite electrode with deformable primer layer for boosting fast-charging capability of lithium-ion batteries** — score: 0.400 The article presents a novel design for graphite electrodes used in lithium-ion batteries, featuring a porosity-gradient and a deformable primer layer. This innovative structure aims to enhance the fast-charging capabilities of the batteries, addressing a critical challenge in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Internal resistance and impedance act as cross-scale indicators of lifespan-safety-performance for lithium-ion batteries** — score: 0.400 The article explores how internal resistance and impedance can serve as crucial indicators across different scales for evaluating the lifespan, safety, and performance of lithium-ion batteries. By examining these parameters, the authors aim to enhance the understanding of battery behavior and inform better management and design practices in energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Spontaneous Nanostructuring of Extended π -Conjugated Carbonyls Enables High-Capacity, Long-Life, and Fast-Charging Cathodes for Sodium-Ion Batteries** — score: 0.400 The article discusses the development of high-capacity and long-life cathodes for sodium-ion batteries, achieved through the spontaneous nanostructuring of extended π -conjugated carbonyl compounds. This innovative approach not only enhances the battery's energy storage capacity but also significantly improves its charging speed and overall performance. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Improving the electro-chemo-mechanical stability of nickel-rich cathodes through tungsten modification for high-performance solid-state batteries** — score: 0.400 The article discusses the enhancement of nickel-rich cathodes' electro-chemo-mechanical stability through the incorporation of tungsten, aimed at optimizing their performance in solid-state batteries. The findings, published in *Energy Storage Materials*, focus on improving longevity and efficiency in high-performance battery applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Enhancing fast-charging capability of silicon-based lithium-ion batteries via an ionically conductive binder** — score: 0.400 This article discusses the development of a new ionically conductive binder that significantly improves the fast-charging performance of silicon-based lithium-ion batteries. The research highlights the binder's ability to enhance ion transport and overall battery efficiency, addressing critical challenges in commercializing high-capacity silicon anodes. Journal: *ScienceDirect Publication: Nano Energy*
- **A dilute nitrate electrolyte for all-climate batteries** — score: 0.400 You et al. present a novel approach utilizing dilute lithium nitrate as a cost-effective electrolyte for lithium-ion batteries. This strategy not only lowers material expenses but also enhances battery stability, facilitating reliable performance across diverse climatic conditions in pouch cell designs. Journal: *Joule*
- **Geopolymer-based electrolytes derived from industrial solid waste for structural energy storage** — score: 0.400 The study presents a novel approach to utilizing industrial solid waste, specifically fly ash and slag, to create a geopolymer-based electrolyte that enhances zinc-ion brick batteries. The engineered material not only offers mechanical strength and durability but also facilitates efficient ion conduction, demonstrating potential for dual-function structural components that support loads while providing energy storage capabilities. Journal: *Joule*

- **Carbon nanotube-based ternary-additive restructured hydrogel electrolyte for high-performance flexible Zn-air batteries** — score: 0.400 The article presents a novel hydrogel electrolyte combining carbon nanotubes and ternary additives, which significantly enhances the performance of flexible Zn-air batteries. This advancement demonstrates improved efficiency and stability, promising advancements in energy storage technologies. Journal: *ScienceDirect Publication: Nano Energy*
- **Polyether electrolyte with selective polymerization of epichlorohydrin for monomer solvated Li⁺ transport and stabilized interfaces** — score: 0.300 The article discusses the development of a polyether electrolyte designed for enhanced lithium ion (Li⁺) transport through selective polymerization of epichlorohydrin. This innovation aims to create more stable interfaces within energy storage systems, thereby improving overall performance and reliability. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Activating ideal solid solution behavior via Mn energy level regulation toward 10C-level layered sodium oxides** — score: 0.300 The article discusses a method for enhancing the performance of 10C-level layered sodium oxides by regulating the energy levels of manganese (Mn) within the material. This approach aims to achieve ideal solid solution behavior, which can potentially improve energy storage capabilities. The findings contribute valuable insights into the optimization of sodium-based materials for energy applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Single-phase gradient-solvation-electrolyte-stabilized Li metal batteries** — score: 0.300 The article discusses the development of lithium metal batteries enhanced by a gradient-solvation approach, which incorporates a specific ligand anti-solvent into an anion-rich ether electrolyte. This innovation results in significant improvements in the battery's cycle life, energy density, and capacity retention. Journal: *Nature*
- **Defect-rich microenvironments optimize bulk doping for upcycling spent Ni-rich cathodes** — score: 0.200 The study investigates how defect-rich microenvironments can enhance bulk doping processes to effectively recycle spent nickel-rich cathodes from batteries. This approach aims to improve material performance and sustainability in energy storage applications, highlighting advancements in the upcycling of battery components. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **The “concentration-purification-delivery” mechanism enabled by silanized attapulgite achieves ultra-stable Zn metal anodes** — score: 0.200 The article discusses a novel “concentration-purification-delivery” mechanism utilizing silanized attapulgite to enhance the stability of zinc metal anodes, significantly improving their performance in energy storage applications. The research highlights the material's potential to revolutionize zinc-based batteries through enhanced electrochemical properties. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Prelithiation strategies for lithium-ion batteries: From component-specific design to scalable manufacturing** — score: 0.200 The article discusses various prelithiation strategies for lithium-ion batteries, emphasizing the importance of component-specific designs and their scalability for manufacturing. It aims to improve battery performance and longevity by exploring innovative methods that enhance lithium availability during the charging process. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Strain-Adaptive Aluminum Fluoride Armor Engineering for Durable Silicon-Carbon Anodes with Suppressed Swelling** — score: 0.200 This article explores the development of strain-adaptive aluminum fluoride armor designed to enhance the durability of silicon-carbon anodes in energy storage applications. The innovative engineering approach aims to significantly suppress swelling issues typically associated with these anodes, thereby improving their performance and longevity in energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Kinetic-Thermodynamic Balance-Driven Design of Bromine Complexants toward High-Performance Zn-Br₂ Batteries** — score: 0.200 The article discusses the design of bromine complexants aimed at enhancing the performance of zinc-bromine batteries by balancing kinetic and thermodynamic factors. The authors explore innovative approaches to optimize the efficiency and energy

storage capabilities of these batteries, which are gaining interest for their potential in large-scale energy applications. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **A dual-site phosphorus-based acceptor with hydrogen bond donor enables theoretically high-capacity F-ion batteries based on unusual Cu³⁺ fluorides** — score: 0.200 The article discusses the development of a novel phosphorus-based dual-site acceptor that incorporates hydrogen bond donors to enhance fluoride-ion battery performance. The research focuses on the unique properties of Cu³⁺ fluorides, showcasing their potential to achieve significantly high energy capacities in next-generation batteries. Journal: *ScienceDirect Publication: Nano Energy*
- **Stabilizing V-based cathode via a medium-entropy strategy for high-performance zinc-ion batteries** — score: 0.200 The article discusses a medium-entropy strategy employed to stabilize vanadium-based cathodes, enhancing the performance of zinc-ion batteries. The research highlights the improved energy efficiency and operational stability facilitated by this method, indicating significant potential for future battery technologies. Journal: *ScienceDirect Publication: Nano Energy*
- **Amorphous-MOF-derived heterojunctions with continuous interfaces for high-performance lithium-sulfur batteries** — score: 0.200 The article discusses the development of heterojunctions derived from amorphous metal-organic frameworks (MOFs) with continuous interfaces, which are designed to enhance the performance of lithium-sulfur batteries. These advancements aim to improve energy storage efficiency and cycle life, showcasing significant potential for future battery technologies. Journal: *ScienceDirect Publication: Nano Energy*
- **Effect of coating type on the structure and electrochemical performance of silicon anodes for lithium-ion batteries** — score: 0.200 The study investigates how different coating types influence the structural integrity and electrochemical performance of silicon anodes used in lithium-ion batteries. The findings suggest that specific coatings enhance the anode's performance, potentially leading to improvements in battery efficiency and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **Dual-Network Eutectogel Electrolyte with Enhanced Interfacial Stability for High-Performance Flexible Zinc-Air Batteries** — score: 0.200 The article discusses the development of a dual-network eutectogel electrolyte designed to improve interfacial stability in flexible zinc-air batteries, potentially enhancing their performance. The authors highlight the innovative contribution of this materials science advancement to energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Silicon embedded into graphite by van der Waals force for high-performance Li-ion batteries** — score: 0.200 This article discusses a novel method of embedding silicon within graphite using van der Waals forces to enhance the performance of lithium-ion batteries. The authors present findings that suggest this approach could improve energy density and overall battery efficiency, paving the way for advanced energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Molecular Barrier Engineering Against Cryogenic Solvent Aggregation Enables Sodium-Ion Battery Operation at -60°C** — score: 0.100 This article discusses the development of molecular barrier engineering techniques to mitigate cryogenic solvent aggregation, which significantly enhances the performance of sodium-ion batteries at extremely low temperatures, specifically down to -60°C. The innovations presented aim to improve battery efficiency and reliability in cold environments, addressing a critical challenge in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Academic battery passports for the regulatory era of energy storage** — score: 0.100 The article discusses the increasing importance of battery passports for ensuring traceability and safety in energy storage. It critiques the current narrative-driven approach of academic battery research and introduces a framework for creating interoperable, machine-readable battery passports that can enhance regulation and inform artificial intelligence and life cycle assessments in the field. Journal: *Joule*
- **From cloning to gene-editing: the enduring legacy of Dolly the sheep** — score: 0.000 The

article reflects on Dolly the sheep’s significance as a pioneering figure in cloning and how her legacy informs contemporary debates surrounding reproductive technologies and genetic engineering. It emphasizes the societal implications of such advancements, suggesting that Dolly’s story offers valuable insights for ongoing discussions in the field. Journal: *Nature*

- **Universities are relying on AI-detection software to catch cheating. How well do the programs work?** — score: 0.000 The article explores the effectiveness of AI-detection software being utilized by universities to identify cheating through AI-generated content. It highlights the differing methodologies and effectiveness of various technology companies’ tools, raising questions about their reliability in academic integrity enforcement. Journal: *Nature*
- **Deep-sea oddities and boatloads of other new species — June’s best science images** — score: 0.000 The article showcases a selection of striking scientific photographs from June, highlighting the discovery of unique deep-sea species alongside a variety of other newly identified organisms, as curated by Nature’s photography team. Journal: *Nature*
- **The rise of computer chips — and the race to control them** — score: 0.000 The article examines the historical development of computer chips, highlighting their significance in the evolution of computing technologies. It shifts the focus from traditional narratives centered on software and tech entrepreneurs in California to emphasize the critical role of hardware in shaping the computing landscape. Journal: *Nature*
- **Time to give hydration breaks the red card? What science says about keeping cool** — score: 0.000 The article discusses the misuse of cooling breaks in sports, highlighting how their improper application is damaging the credibility of heat-health research. It suggests a reevaluation of hydration strategies to ensure athlete safety while maintaining scientific integrity. Journal: *Nature*
- **Daily briefing: How cooperation built the world** — score: 0.000 The article by Rowan Hooper highlights the often underappreciated role of collaboration in nature, contrasting it with the prevailing focus on competition. It also delves into the current state of trust in science, implying a need for reassessment in how we value cooperative behaviors in both natural ecosystems and scientific communities. Journal: *Nature*
- **Temporal potential-induced regulation for extending alkaline electrolyzer lifespan under power fluctuations** — score: 0.000 Xia et al. introduce a novel method to enhance the durability of large-scale alkaline water electrolyzers by optimizing power supply fluctuations, all without altering existing catalyst or electrolyzer designs. This innovative approach effectively extends the lifespan of electrolyzers, offering a scalable solution that connects electrical engineering with electrochemistry and supports the integration of renewable energy into hydrogen production systems. Journal: *Joule*
- **Electrical double-layer dynamics regulation for maximized electrostatic energy harvesting of liquid-solid triboelectric nanogenerators** — score: 0.000 The article discusses advancements in liquid-solid triboelectric nanogenerators, emphasizing a switch-enhanced generator that boosts energy output and adaptability by addressing key limitations in energy harvesting. This innovation facilitates efficient electricity generation from water-solid friction across various applications, ultimately serving as a sustainable power source for electronic devices. Journal: *Joule*
- **Engineered living materials need engineered EU regulation** — score: 0.000 The article by Kai Purnhagen and Justus Wesseler argues for the need to reform EU regulations to facilitate the market entry of engineered living materials, highlighting current legal barriers that hinder their development and adoption. Journal: *Nature Materials*